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Engineering Thesis – Working Draft

Flight Dynamics Modelling of a Boeing 777-like Aircraft and Autopilot Design Using Nonlinear Model Predictive Control

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Abstract

This engineering thesis concerns the development of a nonlinear flight dynamics simulation environment for a Boeing 777-like transport aircraft and the design of an autopilot concept based on Nonlinear Model Predictive Control. The aircraft is treated as a large twin-engine wide-body transport configuration inspired by publicly available Boeing 777-300ER data, but the model is not intended to reproduce a certified manufacturer simulation model. The work therefore focuses on engineering consistency, transparent assumptions, and a modular software structure suitable for control design studies.

The thesis describes the adopted coordinate systems, state and input definitions, mass and inertia estimates, standard atmosphere model, first-cut aerodynamic model, propulsion assumptions, trimming requirements, and validation strategy. The implementation is organised as a MATLAB/Simulink-oriented project with reusable data modules, feature tests and reproducible data-processing tools. Particular attention is given to the separation between fixed aircraft properties, such as mass and inertia, and force quantities that depend on environmental assumptions such as gravitational acceleration.

The expected result of the project is a coherent simulation plant that can be trimmed in representative flight conditions and used as a basis for longitudinal and lateral-directional NMPC autopilot modes. The proposed controller architecture is designed to handle input limits, actuator rate limits, state constraints, and trajectory tracking objectives within a finite-horizon optimisation problem.

Keywords: flight dynamics, six-degree-of-freedom model, Boeing 777-like aircraft, nonlinear model predictive control, autopilot, MATLAB, Simulink.

Model Status and Data Notice

This document is a working engineering-thesis draft developed in parallel with the software implementation. The aircraft referred to as Boeing 777-like is a simplified engineering model of a large twin-engine transport aircraft. Its geometry, mass properties and propulsion assumptions are inspired by publicly available information about the Boeing 777 family, but the model is not an official Boeing model, does not use proprietary manufacturer data and must not be used for certification, operational planning or safety-critical analysis.

All numerical values that are not directly available from public reference material are treated as first-cut engineering estimates. These estimates are intended to support early integration of the flight dynamics model, trimming routines and control algorithms. They should be refined through validation, sensitivity studies and comparison with credible public performance data.

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Nomenclature

Symbol	Meaning
$\boldsymbol{r}_n$	position vector expressed in the navigation frame
$\boldsymbol{v}_b$	velocity vector expressed in the body frame
$\boldsymbol{\omega}_b$	angular velocity vector expressed in the body frame
ϕ, θ, ψ	Euler roll, pitch and yaw angles
p, q, r	body-axis roll, pitch and yaw rates
α	angle of attack
β	sideslip angle
V	true airspeed
q_∞	dynamic pressure
S	wing reference area
b	wing span
$\bar{c}$	mean aerodynamic chord
C_L, C_D, C_Y	lift, drag and side-force coefficients
C_l, C_m, C_n	rolling, pitching and yawing moment coefficients
δ_e	elevator deflection
δ_a	effective aileron deflection
δ_r	rudder deflection
T_1, T_2	left and right engine thrust
m	aircraft mass
$\boldsymbol{J}$	inertia matrix about the centre of gravity
g_0	standard gravitational acceleration used as the current project default

Abbreviation	Meaning
6-DOF	six-degree-of-freedom
CG	centre of gravity
ISA	International Standard Atmosphere
LNAV	lateral navigation

MAC	mean aerodynamic chord
MPC	model predictive control
NMPC	nonlinear model predictive control
NED	North-East-Down navigation frame
VNAV	vertical navigation

Chapter 1

Introduction

Automatic flight control systems are a fundamental part of modern transport aircraft. Their role extends beyond basic stabilisation: they reduce pilot workload, improve flight-path tracking, support energy management and help maintain operation within structural, aerodynamic and actuator constraints. The control problem is particularly demanding for large transport aircraft because of their high mass, large inertia, coupled longitudinal and lateral-directional dynamics, altitude-dependent performance and strict actuator limitations.

Classical autopilot architectures are often organised as hierarchical control systems. Inner loops stabilise angular rates or attitude variables, while outer loops generate commands for altitude, airspeed, heading or path tracking. This structure is effective and widely used, but it also separates tracking objectives from constraints. Nonlinear Model Predictive Control offers a different design philosophy: the nonlinear prediction model, performance objective and operational constraints are treated within one finite-horizon optimisation problem [1, 2].

The subject of this thesis is the development of a Boeing 777-like flight dynamics simulator and an NMPC-based autopilot concept. The phrase Boeing 777-like denotes an educational model of a large twin-engine wide-body aircraft whose dimensions and mass envelope are qualitatively close to the Boeing 777-300ER class. It does not denote an exact aircraft model. This distinction is important because accurate aerodynamic, engine and flight-control data for commercial aircraft are not publicly available in the detail required for a certified simulator.

1.1 Engineering Motivation

The project combines several areas of aerospace engineering: rigid-body flight mechanics, coordinate-frame conventions, aerodynamic modelling, propulsion modelling, numerical simulation, trimming and constrained control design. The value of the project lies in building the complete modelling chain. Instead of studying the controller in isolation, the thesis develops a simulation plant, defines validation criteria and then uses the plant as the basis for autopilot synthesis.

The project also has a software-engineering motivation. A flight dynamics model becomes difficult to validate when geometry, mass data, atmosphere assumptions, aerodynamic coefficients and actuator limits are scattered through ad hoc files. For that reason, the implementation is organised into reusable MATLAB modules, with explicit units and feature-oriented checks. This makes the model easier to review and reduces the risk of hidden convention errors.

1.2 Problem Statement

The main engineering problem considered in this thesis can be stated as follows:

> How can a coherent nonlinear flight dynamics model of a large transport aircraft be constructed from public and estimated data, and how can it be used as the basis for an NMPC autopilot that respects actuator, thrust and flight-envelope constraints?

This problem requires both modelling and control-design decisions. The simulation plant must be detailed enough to represent the dominant aircraft dynamics, but simple enough to be implemented, trimmed and validated within the scope of an engineering thesis. The NMPC prediction model must be computationally tractable while remaining dynamically consistent with the plant in the operating region of interest.

1.3 Thesis Structure

Chapter 2 defines the aim, scope and success criteria of the project. Chapter 3 introduces the theoretical background required for flight dynamics modelling and predictive control. Chapter 4 presents the mathematical model of the aircraft. Chapter 5 describes the software implementation. Chapter 6 discusses trimming and validation. Chapter 7 presents the planned NMPC autopilot architecture. Chapter 8 defines the simulation study programme. Chapter 9 summarises the current state of the project and outlines further work.

Chapter 2

Aim and Scope of the Thesis

2.1 Main Aim

The main aim of this thesis is to develop a nonlinear simulation environment for a Boeing 777-like transport aircraft and to design an autopilot concept based on Nonlinear Model Predictive Control. The simulator is intended to support analysis of six-degree-of-freedom aircraft motion, trimming in representative flight conditions and evaluation of constrained control algorithms.

2.2 Detailed Objectives

The detailed objectives of the work are:

1. to define coordinate frames, sign conventions, state vectors and input vectors used throughout the project,
2. to prepare a first-cut geometry, mass and inertia model of a Boeing 777-like aircraft,
3. to implement a simple International Standard Atmosphere model and fixed project constants,
4. to implement a simplified aerodynamic coefficient model suitable for early flight-dynamics integration,
5. to define a propulsion and actuator modelling approach,
6. to formulate the nonlinear six-degree-of-freedom equations of motion,
7. to prepare trimming and validation procedures for selected reference cases,
8. to design longitudinal and lateral-directional NMPC control structures,
9. to evaluate the model and controller using repeatable simulation scenarios.

2.3 Scope

The scope of the thesis includes rigid-body flight dynamics, simplified aerodynamics, simplified propulsion, actuator limits, standard atmosphere modelling, trimming, qualitative validation and NMPC autopilot design. The implementation target is MATLAB/Simulink, with MATLAB data modules used to define reusable aircraft and environment parameters.

The following topics are outside the scope of the present work:

- structural flexibility and aeroelastic effects,
- detailed hydraulic, electrical or avionics system models,
- certified flight-control laws,
- proprietary aerodynamic or engine data,
- high-fidelity pilot-display or flight-management-system simulation,
- certification-level validation.

2.4 Assumptions and Limitations

The aircraft is treated as a rigid body. The Earth is initially approximated as flat and non-rotating, and the navigation frame is a local North-East-Down frame. All internal calculations are performed in SI units. The current implementation uses fixed standard gravity $g_0 = 9.806\,65\,\mathrm{m/s^2}$ as a project-wide constant. Gravity-dependent force quantities are computed from mass and the selected gravity value rather than stored as immutable mass properties.

The aerodynamic model is intentionally simple. Its purpose is to provide physically reasonable signs, magnitudes and coupling terms for early simulation and control-design studies. The model is expected to be refined after trim tests and dynamic-response validation.

2.5 Success Criteria

The project will be considered successful if the following criteria are met:

1. the six-degree-of-freedom plant can be executed in simulation without numerical instability caused by modelling errors,
2. trim points can be obtained for several representative flight conditions,

3. open-loop dynamic responses are qualitatively consistent with expectations for a large transport aircraft,
4. the NMPC controller can stabilise and track selected flight variables,
5. actuator, thrust and selected state constraints are respected during simulation,
6. the obtained results are documented with plots, tables and engineering interpretation.

Chapter 3

Theoretical Background

3.1 Reference Frames in Flight Mechanics

Aircraft motion is described using several coordinate frames [3, 4]. The navigation frame used in this project is a local North-East-Down frame. Its axes point north, east and down, respectively. Altitude is therefore related to the down coordinate by $h = -D$.

The body frame is fixed to the aircraft. Its x_b axis points forward through the nose, its y_b axis points through the right wing and its z_b axis points downward. This convention is right-handed and is compatible with many aerospace simulation formulations. The wind frame is aligned with the air-relative velocity vector and is convenient for defining lift, drag, angle of attack and sideslip.

3.2 Six-Degree-of-Freedom Aircraft Motion

A six-degree-of-freedom aircraft model describes translational and rotational motion of a rigid body. The translational dynamics are obtained from Newton's second law expressed in a rotating body frame. The rotational dynamics are obtained from Euler's rigid-body equations. The state typically contains position, velocity, attitude and angular rates. In this project the state is extended by engine thrust and actuator states so that propulsion and control-surface dynamics are not assumed to be instantaneous.

3.3 Aerodynamic Coefficient Modelling

Aerodynamic forces and moments are commonly represented using nondimensional coefficients. Dynamic pressure is defined as

$$q_\infty = \frac{1}{2}\rho V^2, \tag{3.1}$$

where ρ is air density and V is true airspeed. Lift, drag and moments are then scaled using reference area, span and mean aerodynamic chord. For example,

$$L = q_\infty S C_L, \qquad D = q_\infty S C_D, \qquad M = q_\infty S \bar{c} C_m. \tag{3.2}$$

For an engineering first cut, the coefficients may be approximated by linear derivatives and simple nonlinear drag terms. Such a model is not sufficient for high-fidelity prediction near stall or in complex configurations, but it is adequate for early integration and control-design experiments when its validity range is clearly stated.

3.4 Propulsion and Actuator Dynamics

The propulsion model maps throttle commands to thrust. A realistic engine does not change thrust instantaneously, therefore the simulation model should include at least first-order spool dynamics. Similarly, control surfaces should include position limits, rate limits and possibly first-order actuator dynamics. These elements are important because NMPC relies on accurate constraint representation.

3.5 Nonlinear Model Predictive Control

Nonlinear Model Predictive Control solves an optimisation problem at each control update. The controller predicts future system behaviour over a finite horizon and selects an input sequence that minimises a cost function while satisfying constraints [1, 2]. Only the first input is applied to the plant, and the optimisation is repeated at the next sampling instant.

NMPC is attractive for aircraft autopilot design because it can explicitly include actuator limits, thrust limits, attitude limits, airspeed bounds, load-factor constraints and path-tracking objectives. Its main disadvantage is computational complexity, which motivates the use of reduced-order prediction models and carefully selected sampling times.

Chapter 4

Mathematical Model of the Aircraft

4.1 State Vector

The full nonlinear plant state is defined consistently with the project convention file as

$$\boldsymbol{x} = \begin{bmatrix} N & E & D & u & v & w & \phi & \theta & \psi & p & q & r & T_1 & T_2 & \delta_e & \delta_a & \delta_r \end{bmatrix}^T. \tag{4.1}$$

The variables N, E and D define position in the navigation frame. The variables u, v and w are body-axis velocity components. The Euler angles are roll ϕ, pitch θ and yaw ψ. The angular rates p, q and r are expressed in the body frame. The remaining states represent left and right engine thrust and actual control-surface deflections.

4.2 Command Input Vector

The command input vector is

$$\boldsymbol{u}_{cmd} = \begin{bmatrix} \delta_{e,cmd} & \delta_{a,cmd} & \delta_{r,cmd} & \tau_{1,cmd} & \tau_{2,cmd} \end{bmatrix}^T, \tag{4.2}$$

where $\delta_{e,cmd}$, $\delta_{a,cmd}$ and $\delta_{r,cmd}$ are commanded elevator, effective aileron and rudder deflections. The throttle commands $\tau_{1,cmd}$ and $\tau_{2,cmd}$ are nondimensional values in the range from zero to one.

4.3 Kinematics

The position kinematics are written as

$$\dot{\boldsymbol{r}}_n = \boldsymbol{C}_{nb}\boldsymbol{v}_b, \tag{4.3}$$

where $\boldsymbol{C}_{nb}$ transforms a vector from the body frame to the navigation frame. The Euler angle rates are related to body rates by

$$\begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} = \begin{bmatrix} 1 & \sin\phi\tan\theta & \cos\phi\tan\theta \\ 0 & \cos\phi & -\sin\phi \\ 0 & \sin\phi/\cos\theta & \cos\phi/\cos\theta \end{bmatrix} \begin{bmatrix} p \\ q \\ r \end{bmatrix}. \tag{4.4}$$

This representation is simple and appropriate for normal aircraft attitudes, but it becomes singular as θ approaches $\pm 90°$.

4.4 Translational Dynamics

The translational equations in the rotating body frame are

$$m\dot{\boldsymbol{v}}_b = \boldsymbol{F}_b - m\boldsymbol{\omega}_b \times \boldsymbol{v}_b, \tag{4.5}$$

where

$$\boldsymbol{v}_b = \begin{bmatrix} u & v & w \end{bmatrix}^T, \qquad \boldsymbol{\omega}_b = \begin{bmatrix} p & q & r \end{bmatrix}^T. \tag{4.6}$$

The total body-frame force is decomposed as

$$\boldsymbol{F}_b = \boldsymbol{F}_{aero,b} + \boldsymbol{F}_{eng,b} + \boldsymbol{F}_{g,b}. \tag{4.7}$$

The gravitational force is computed using the current gravity assumption. At the present stage the project uses the fixed standard value g_0, defined in the constants module.

4.5 Rotational Dynamics

The rotational equations are

$$\boldsymbol{J}\dot{\boldsymbol{\omega}}_b + \boldsymbol{\omega}_b \times (\boldsymbol{J}\boldsymbol{\omega}_b) = \boldsymbol{M}_b, \tag{4.8}$$

where $\boldsymbol{J}$ is the inertia matrix about the centre of gravity and $\boldsymbol{M}_b$ is the total body-frame moment. The first-cut inertia model uses

$$\boldsymbol{J} = \begin{bmatrix} I_{xx} & 0 & -I_{xz} \\ 0 & I_{yy} & 0 \\ -I_{xz} & 0 & I_{zz} \end{bmatrix}. \tag{4.9}$$

The current implementation sets $I_{xz} = 0$ to keep the early model simple. This assumption should be revisited after the plant can be trimmed and linearised.

4.6 Mass and Inertia Model

The mass module defines the mass envelope, nominal cruise mass, nominal approach mass and estimated inertia tensor. Because public certified inertia data for the aircraft are not available, the inertia tensor is estimated using effective radii of gyration based on public aircraft dimensions. This method is suitable for early integration but must be treated as an engineering approximation.

The implementation separates mass properties from gravity-dependent forces. The function `b777_mass_inertia.m` returns a structure containing mass and inertia values, while force quantities are obtained by evaluating a function handle:

```
mass = b777_mass_inertia();
forces = mass.forces(mass.standard_gravity_mps2);
```

Listing 4.1: Current gravity-dependent mass force interface.

This design allows the project to use fixed standard gravity at the current stage while preserving a clean interface for a future variable-gravity environment model.

4.7 Atmosphere Model

The atmosphere model is a simplified International Standard Atmosphere implementation up to 20 km. It returns temperature, pressure, density, speed of sound and viscosity. The same standard gravity constant used by the mass module is used in the hydrostatic pressure relations. This avoids hidden inconsistencies between atmosphere and mass-force calculations.

4.8 Aerodynamic Model

The aerodynamic model was built as a staged, source-oriented database rather than as a single analytical coefficient block. This choice was made because a Boeing 777-class aircraft cannot be represented credibly by one simplified polar while still retaining the traceability required for trimming, linearisation and NMPC design. The implemented interface is provided by `b777_aero_database.m`. It separates the plant-facing coefficient request from the source, status and confidence level of each coefficient group.

4.8.1 Development Objective and Data Strategy

The main difficulty is that detailed Boeing 777 aerodynamic tables are not public. The model is therefore deliberately a B777-like engineering model, not a certified Boeing model. The development strategy is to combine public transport-aircraft aerodynamic data with semi-empirical derivative estimates and then accept data into the active plant only after validation gates have been passed.

The adopted source hierarchy is:

- B777-like geometry, reference area, span, mean aerodynamic chord and centre-of-gravity data from the project aircraft data modules,
- NASA Common Research Model data for the clean-cruise longitudinal baseline [5, 6, 7],
- Digital DATCOM for missing static, damping and control derivative candidates [8, 9, 10],
- CRM-HL and DATCOM-ready high-lift increments for approach and landing development [11],
- OpenFOAM as a later verification tool for selected cases, not as the primary database source [12].

4.8.2 Development Pipeline

Table 4.1 summarises the construction path used in the project. The important design rule is that source data, candidate data and active plant data are not treated as the same thing.

Table 4.1: Aerodynamic database construction pipeline.

Layer	Engineering role and status
Geometry and mass	B777-like project geometry and mass properties define reference area, span, mean aerodynamic chord and CG for force and moment scaling. Active.
Clean cruise	NASA CRM NTF197 TWICS data provide the active clean C_L, C_D and C_m baseline inside the committed transonic cruise range.
Analytical seed	`b777_aero_simple.m` provides fallback coefficients and low-confidence residual terms when validated table data are not available. Active as fallback.
Static and damping derivatives	Digital DATCOM Mach-grid tables provide source-backed derivative candidates. Not promoted until validation and warning review are complete.
Control derivatives	Digital DATCOM elevator and aileron control-grid tables provide direct control-effectiveness candidates and document direct coverage gaps. Candidate layer.
Derived gap-fill	Geometry-based values derived from DATCOM candidates cover active-interface gaps such as selected rudder terms and lateral drag increments. Candidate layer.
High-lift increments	A CRM-HL/DATCOM-ready clean, approach and landing table adds configuration increments and $C_{L,\max}$ metadata. Active as an engineering seed.
Validation gates	Trim, modal and later performance validation evidence control whether candidate data may replace the active seed layer.

4.8.3 Clean-Cruise Baseline

At the current implementation level, the active clean-cruise longitudinal table is a Mach-alpha grid derived from NASA CRM NTF Test 197 TWICS-corrected force and moment data. The selected clean candidate is identified in the source files by `CONFIG=3`, `CONFTS=1` and `CONFT=0`. The table stores C_L, C_D and C_m over $M \approx 0.70$ to 0.87 and $\alpha = -2^\circ$ to 9°. Linear interpolation in Mach number and angle of attack is used only inside this committed range.

The CRM geometry is not interpreted as the Boeing 777 geometry. Instead, CRM is used as a public, well documented transport-aircraft aerodynamic baseline. The B777-like reference geometry is used later when the nondimensional coefficients are converted into forces and moments.

4.8.4 Residual Derivative and Control Layers

The clean CRM grid does not provide the complete coefficient set required by a six-degree-of-freedom plant. Residual effects are therefore added through static, damping and control derivative layers. In compact form, the assembled coefficient vector can be written as

$$\boldsymbol{C}_{aero} = \boldsymbol{C}_{clean}(M, \alpha) + \Delta\boldsymbol{C}_{\beta}(\beta) + \Delta\boldsymbol{C}_{rates}(\hat{p}, \hat{q}, \hat{r}) + \Delta\boldsymbol{C}_{ctrl}(\delta_e, \delta_a, \delta_r) + \Delta\boldsymbol{C}_{config}, \quad (4.10)$$

where $\boldsymbol{C}_{clean}$ contains the CRM clean longitudinal coefficients, $\Delta\boldsymbol{C}_{\beta}$ contains sideslip effects, $\Delta\boldsymbol{C}_{rates}$ contains damping effects, $\Delta\boldsymbol{C}_{ctrl}$ contains control-surface increments and $\Delta\boldsymbol{C}_{config}$ contains flap, slat and gear increments. The present active derivative values are still low-confidence engineering seeds, but their schema matches the intended replacement path through Digital DATCOM.

A reproducible Digital DATCOM package has been added before activating any DATCOM-derived values. Clean B777-like static and damping decks are run at $M =$ 0.30, 0.50, 0.60, 0.70 and 0.80, and the extracted derivatives are consolidated into a Mach-dependent candidate table. A companion control-grid workflow extracts elevator and aileron effectiveness candidates from `SYMFLP` and `ASYFLP` cases over the same Mach stations. Terms not printed directly by the current DATCOM outputs, including C_{D_β}, C_{Y_r}, rudder derivatives and lateral control drag increments, are stored in a derived gap-fill candidate table. These gap-fill values are computed from the DATCOM candidate derivatives and the B777-like geometry, so they preserve active-interface coverage without being treated as direct DATCOM printouts.

4.8.5 Analytical Seed Model

The first-cut seed model provides lift, drag, side-force and moment coefficients as functions of angle of attack, sideslip, nondimensional angular rates and control-surface deflections. A representative longitudinal form is

$$C_L = C_{L0} + C_{L\alpha}\alpha + C_{Lq}\hat{q} + C_{L\delta_e}\delta_e, \quad (4.11)$$

$$C_m = C_{m0} + C_{m\alpha}\alpha + C_{mq}\hat{q} + C_{m\delta_e}\delta_e, \quad (4.12)$$

with a drag model of the form

$$C_D = C_{D0} + kC_L^2 + C_{D\beta}\beta^2 + C_{D\delta_e}\delta_e^2 + C_{D\delta_a}\delta_a^2 + C_{D\delta_r}\delta_r^2. \quad (4.13)$$

The coefficient signs follow the body-frame and control-surface conventions defined in the project.

The seed model has two roles. First, it provides a simple complete coefficient set for software integration. Second, it defines the fallback behaviour outside the validated CRM grid. Whenever the database uses fallback values, the source metadata indicate that the coefficient group is not coming from validated table data.

4.8.6 Configuration and High-Lift Layer

The database layer also stores configuration metadata for clean, approach and landing cases. The current clean configuration uses the CRM grid only inside its local validity range. Inside that range, the CRM grid supplies the clean longitudinal baseline and the derivative table supplies residual lateral-directional, damping and control effects.

The approach and landing configurations use a separate CRM-HL/DATCOM-ready high-lift seed table with explicit flap, slat, gear, lift-increment, drag-increment, pitching-moment-increment and $C_{L,\max}$ metadata. The $C_{L,\max}$ value is currently exposed for trim and validation checks, not as a complete stall model. These values are sufficient for software integration and initial trim testing, but they are not intended to represent certified aircraft data.

4.8.7 Candidate Promotion Policy

Static, control and damping derivatives are exposed through dedicated tables so that DATCOM or validation-derived estimates can later replace individual groups without changing the plant interface. The current DATCOM Mach-grid, control-grid and derived gap-fill files are therefore treated as review artifacts rather than active aerodynamic tables.

The DATCOM candidates are controlled by automatic promotion gates. The gates check source identifiers, case traceability, run quality, Mach-grid consistency, coefficient signs, confidence thresholds, control-derivative coverage and active-interface coverage. Clean-cruise longitudinal trim validation and restricted longitudinal modal validation have passed for three DATCOM-candidate cases, but the candidate derivatives remain inactive until the warning review is complete. This prevents a partially reviewed candidate table from silently entering the nonlinear plant. The final aerodynamic database should be treated as acceptable for NMPC studies only after trim, modal and performance validation gates have been passed.

4.8.8 Conversion to Forces and Moments

The plant-facing aerodynamic load model is implemented by `b777_aero_forces_moments.m`. It evaluates the coefficient database, computes dynamic pressure, transforms the wind-axis force vector into the body frame and scales the moment coefficients by the project reference span and mean aerodynamic chord:

$$\boldsymbol{F}_w = q_\infty S \begin{bmatrix} -C_D & C_Y & -C_L \end{bmatrix}^T, \tag{4.14}$$

$$\boldsymbol{F}_b = \boldsymbol{C}_{bw}(\alpha, \beta)\boldsymbol{F}_w, \tag{4.15}$$

$$\boldsymbol{M}_{ref,b} = q_\infty S \begin{bmatrix} bC_l & \bar{c}C_m & bC_n \end{bmatrix}^T. \tag{4.16}$$

If the aerodynamic reference point is not coincident with the current centre of gravity, the moment is shifted using

$$\boldsymbol{M}_{cg,b} = \boldsymbol{M}_{ref,b} + \boldsymbol{r}_{ref/cg,b} \times \boldsymbol{F}_b. \tag{4.17}$$

This step makes the aerodynamic database usable by the nonlinear plant while keeping the moment-reference policy explicit.

4.9 Actuator and Engine Model

The plant will apply command signals through actuator and engine dynamics. A generic first-order actuator model can be expressed as

$$\dot{\delta} = \mathrm{sat}_{\dot{\delta}_{min},\dot{\delta}_{max}} \left(\frac{\delta_{cmd} - \delta}{\tau_\delta} \right), \tag{4.18}$$

with position limits

$$\delta_{min} \leq \delta \leq \delta_{max}. \tag{4.19}$$

Engine thrust dynamics can be represented analogously by a first-order lag between commanded and actual thrust. These dynamics are essential for a realistic NMPC formulation because they define the achievable rate of change of aircraft forces and moments.

Chapter 5

Implementation

5.1 Project Organisation

The project is organised as a modular MATLAB/Simulink workspace. Aircraft data, environment constants, aerodynamic coefficients and reference cases are stored separately from plant dynamics and controller code. This structure supports incremental validation and makes it easier to replace simplified modules with more accurate versions later in the project.

Table 5.1: Main project directories.

Directory	Purpose
`data`	aircraft, atmosphere, aerodynamic and reference-case data
`plant`	nonlinear six-degree-of-freedom plant implementation
`nmpc`	predictive controller models and controller-generation scripts
`guidance`	mode logic and trajectory-reference generation
`validation`	trim, dynamic-response and autopilot validation scripts
`models`	Simulink model files
`tests`	feature-oriented MATLAB checks
`tools`	DATCOM extraction, validation and reporting helpers
`docs`	documentation and thesis sources

5.2 Data Modules

The current implementation contains the following data modules:

- `b777_constants.m`, which defines project-wide constants such as standard gravity,
- `b777_geometry.m`, which defines B777-300ER-like geometry and reference dimensions,

- `b777_mass_inertia.m`, which defines mass envelope values and first-cut inertia estimates,
- `b777_atmosphere_isa.m`, which provides a simple ISA atmosphere model,
- `b777_aero_database.m`, which defines the staged aerodynamic database interface and source inventory,
- `b777_aero_clean_crm_seed.m`, which exposes the NASA CRM clean Mach-alpha grid,
- `b777_aero_read_crm_clean_grid_csv.m`, which reads the curated NASA CRM clean grid,
- `b777_aero_read_crm_force_moment_csv.m`, which reads NASA CRM force and moment CSV files,
- `b777_aero_read_derivative_csv.m`, which reads source-oriented derivative tables,
- `b777_aero_read_mach_derivative_csv.m`, which reads source-oriented Mach-dependent derivative candidate tables,
- `b777_aero_read_high_lift_csv.m`, which reads source-oriented high-lift increment tables,
- `b777_aero_derivative_seed.m`, which exposes the DATCOM-ready static, control and damping residual derivatives,
- `b777_aero_high_lift_seed.m`, which exposes the CRM-HL/DATCOM-ready clean, approach and landing configuration increments,
- `b777_aero_simple.m`, which defines a first-cut aerodynamic coefficient model,
- `b777_configs.m`, which defines aircraft configurations,
- `reference_cases.m`, which defines representative flight conditions for trim and validation,
- `conventions.m`, which records units, frames, states and input conventions.

The data modules use explicit field names containing units where practical. This convention is intentionally verbose, but it reduces ambiguity and supports code review. For example, mass values are stored with the suffix `_kg`, while speeds use `_mps`.

5.2.1 Aerodynamic Database Build Workflow

The aerodynamic database was implemented as a sequence of reproducible data preparation steps. The workflow is intentionally conservative: each generated candidate table carries a source identifier and remains separate from the active plant table until validation and review criteria are satisfied.

The practical build sequence is:

1. define the B777-like reference geometry and mass properties used for aerodynamic scaling;
2. read the curated NASA CRM clean Mach-alpha grid and expose it as the active clean-cruise longitudinal baseline;
3. preserve the analytical seed model as a complete fallback model for states and coefficient groups not covered by validated data;
4. generate and run Digital DATCOM clean static and damping decks at the selected Mach stations;
5. parse the DATCOM output, convert printed derivatives to project units and consolidate the per-Mach outputs into a candidate derivative table;
6. generate and run elevator and aileron control-effectiveness decks, then consolidate direct control-derivative candidates;
7. derive low-confidence gap-fill candidates for active-interface terms not printed directly by the current DATCOM workflow;
8. solve clean-cruise longitudinal trim cases for the candidate aerodynamic path;
9. linearise a restricted longitudinal model at the same trim points and check the phugoid and short-period modes;
10. evaluate promotion gates before any candidate derivative replaces the active seed table.

The repository contains the scripts required to reproduce those steps. The script `tools/run_digital_datcom.sh` downloads the public-domain DATCOM archive from PDAS, compiles the legacy Fortran source locally and runs a committed B777-like input deck. The extractor `tools/extract_datcom_derivatives.py` parses the static and dynamic derivative tables, converts derivatives printed per degree to derivatives per radian and records skipped cells such as `NDM`.

The wrapper `tools/run_datcom_mach_grid.sh` runs the committed clean single-condition decks at $M = 0.30$, 0.50, 0.60, 0.70 and 0.80. `tools/build_datcom_mach_gr` id_table.py consolidates the per-Mach candidate files into one Mach-dependent table with a grid-level source identifier and retained per-run source identifiers.

The companion wrapper tools/run_datcom_control_grid.sh generates and runs elevator and aileron control-effectiveness decks over the same Mach grid. Its extractor, tools/extract_datcom_control_derivatives.py, parses the DATCOM high-lift/control-device output and produces source-backed candidates for elevator and aileron derivatives. The helper tools/build_datcom_control_grid_table.py consolidates these control candidates and records which entries are not direct DATCOM printouts. tools/build_datcom_gap_fill_candidates.py then derives low-confidence active-interface candidates for C_{D_β}, C_{Y_r}, rudder derivatives and lateral control drag from the DATCOM candidate tables and the B777-like geometry.

The validation scripts form the final part of the workflow. tools/run_trim_validation.py solves clean-cruise longitudinal trim cases for the DATCOM candidate path and writes both a case table and a report. tools/run_longitudinal_modal_validation.py linearises a restricted longitudinal model around the same trim points and checks the phugoid and short-period modes. Finally, tools/evaluate_datcom_promotion_gates.py evaluates whether the direct and derived candidate tables are allowed to replace the active derivative seed. The resulting candidate CSV files, extraction reports, promotion-gate reports, trim-validation artifacts and modal-validation artifacts are stored under data/aerodynamics/raw/datcom, data/aerodynamics/validation and docs. Generated DATCOM run folders are treated as reproducible workspace outputs and are not part of the source repository.

5.3 Startup and Path Management

The file startup.m configures the MATLAB path by adding the project subdirectories required by the current implementation. It avoids relying on the user's current MATLAB path and makes the execution of feature tests repeatable from a clean MATLAB session.

5.4 Feature Tests

The plant directory now includes b777_aero_forces_moments.m, which is the first plant-facing aerodynamic load adapter. It completes the atmosphere-dependent quantities required by the coefficient database, computes dynamic pressure, converts wind-axis lift and drag to body-axis force components, scales aerodynamic moments and applies the configured centre-of-gravity shift.

The project includes feature-oriented MATLAB checks under tests/features. The project-structure test checks the expected top-level folders. The modelling-conventions test checks frame definitions, state-vector sizes and input-vector sizes. The aircraft-data test validates the current geometry, mass, inertia, atmosphere and reference-case data. The aerodynamic-database test checks the staged aerodynamic database interface, source inventory, derivative inventory, CRM clean-grid values, DATCOM-ready residual derivative effects, the reproducible DATCOM Mach-grid, control-grid and derived gap-fill candidate status, trim-validation and modal-validation evidence, CRM-HL/DATCOM-ready high-lift metadata, clean/approach/landing behaviour, aerodynamic force signs and the reference-point-to-CG moment shift. These checks are not substitutes for physical validation, but they catch interface errors early.

5.5 Current Implementation Status

At the current stage, the data layer and the first aerodynamic plant adapter are available. Geometry, mass, inertia, atmosphere and staged first-cut aerodynamics have been defined, including the clean CRM Mach-alpha grid, DATCOM-ready residual derivative metadata, source-backed DATCOM Mach-grid and control-grid derivative candidate run sets, derived gap-fill candidates, clean-cruise longitudinal trim-validation and modal-validation artifacts, CRM-HL/DATCOM-ready high-lift increments and body-axis aerodynamic loads. The next implementation step is to implement engine loads, actuator dynamics and the six-degree-of-freedom equations in MATLAB/Simulink.

Chapter 6

Trimming and Validation

6.1 Role of Trimming

Trimming is a central step in flight dynamics modelling. A trim condition is a steady or quasi-steady operating point in which selected accelerations and tracking errors are zero or match prescribed values. For level cruise, for example, the aircraft should maintain altitude and airspeed with approximately zero angular accelerations and zero flight-path angle.

Without trim points it is difficult to evaluate stability, initialise simulations, linearise the model or design an autopilot. In this project trimming is therefore treated as part of the model-development workflow rather than as a post-processing task. A candidate aerodynamic database is not considered ready for further use merely because its coefficients have plausible signs; it must also allow the aircraft to reach physically reasonable equilibrium conditions.

6.2 Trim Formulation

A trim problem can be expressed as a constrained nonlinear optimisation problem:

$$\min_{\boldsymbol{x},\boldsymbol{u}} \|\boldsymbol{f}_{trim}(\boldsymbol{x},\boldsymbol{u},\boldsymbol{p})\|^2 , \tag{6.1}$$

where $\boldsymbol{x}$ is the state vector, $\boldsymbol{u}$ is the input vector and $\boldsymbol{p}$ contains prescribed flight-condition parameters such as altitude, speed, mass and configuration. The residual vector may include translational accelerations, angular accelerations, flight-path-angle error, sideslip error and actuator equilibrium errors.

6.3 Implemented Longitudinal Trim Check

At the current stage, a restricted clean-cruise longitudinal trim check has been implemented for the DATCOM candidate aerodynamic path. This check does not yet replace the future full six-degree-of-freedom trim solver. Its purpose is narrower: it verifies that the CRM clean baseline combined with the DATCOM elevator candidates can produce realistic cruise equilibrium points before the derivative candidates are considered for promotion.

For a prescribed altitude h, Mach number M, mass m and flight-path angle γ, the speed and dynamic pressure are computed from the ISA atmosphere:

$$V = Ma(h), \tag{6.2}$$

$$q_\infty = \frac{1}{2}\rho(h)V^2. \tag{6.3}$$

For the current level-cruise cases, the attitude relation is approximated by

$$\theta = \alpha + \gamma. \tag{6.4}$$

The elevator deflection is obtained from the zero pitching-moment condition:

$$C_m(M, \alpha, \delta_e) = C_{m,clean}(M, \alpha) + C_{m_{\delta_e}}(M)\delta_e = 0, \tag{6.5}$$

which gives

$$\delta_e = -\frac{C_{m,clean}(M, \alpha)}{C_{m_{\delta_e}}(M)}. \tag{6.6}$$

The remaining scalar unknown is the angle of attack. It is solved from the vertical body-axis force balance:

$$r_z(\alpha) = q_\infty S\left(-C_D \sin\alpha - C_L \cos\alpha\right) + mg\cos\theta = 0. \tag{6.7}$$

After α and δ_e have been found, the required thrust is computed from the axial balance:

$$T = -q_\infty S\left(-C_D \cos\alpha + C_L \sin\alpha\right) + mg\sin\theta. \tag{6.8}$$

In this validation step, thrust is treated as a static-equivalent scalar along the body x axis. Engine pitching moments, actuator states and lateral-directional equilibrium are intentionally outside the scope of this first trim gate.

The implemented script `tools/run_trim_validation.py` evaluates three clean-cruise cases at 10 668 m and 250 000 kg. The case table is written to `data/aerodynamics/validation/trim_validation_cases.csv`, while the summary report is written to `docs/trim_validation_report.md`. The current results are listed in Table 6.1.

Table 6.1: Current clean-cruise longitudinal trim-validation results for the DATCOM candidate path.

Case	Mach	α [deg]	δ_e [deg]	C_L	T [kN]
clean_cruise_m070	0.70010	5.5815	-0.5315	0.6822	143.41
clean_cruise_m075	0.75015	4.4035	0.1550	0.5952	130.21
clean_cruise_m080	0.80000	3.4957	0.9323	0.5238	126.97

The acceptance criteria require the angle of attack to remain inside the clean CRM range and below 8°, the elevator trim to remain within ±5°, the static-equivalent thrust to remain positive and below 30% of the public GE90-115B static thrust estimate, the residual forces to be numerically negligible, the pitching-moment coefficient residual to be near zero and the margin to the current clean $C_{L,\max}$ estimate to remain at least 0.40. All three cases pass these checks, so the validation report records the marker

$$\texttt{DATCOM\_PROMOTION\_TRIM\_VALIDATED=true}. \tag{6.9}$$

This marker does not by itself activate the DATCOM candidate derivatives. It only removes the trim-validation blocker from the promotion gate; the modal gate and DATCOM warning review are evaluated separately.

6.4 Implemented Longitudinal Modal Check

The next validation layer uses the same three clean-cruise trim cases to check the dominant longitudinal modes of the DATCOM candidate path. The implemented script `tools/run_longitudinal_modal_validation.py` forms a restricted longitudinal state vector

$$\boldsymbol{x}_{lon} = \begin{bmatrix} u & w & q & \theta \end{bmatrix}^T. \tag{6.10}$$

The elevator and thrust values are held at their trim values. The aerodynamic model uses the CRM clean C_L, C_D and C_m table, DATCOM candidate C_{L_q} and C_{m_q} derivatives, and the DATCOM elevator effectiveness candidates. The resulting nonlinear residual is linearised numerically around each trim point:

$$\Delta\dot{\boldsymbol{x}}_{lon} = \boldsymbol{A}_{lon}\Delta\boldsymbol{x}_{lon}. \tag{6.11}$$

The finite-difference scheme uses central differences inside the data range and one-sided differences when a trim point lies on a committed Mach-grid boundary. This avoids extrapolating the CRM or DATCOM candidate data.

The current modal-validation results are listed in Table 6.2. The slower pair is classified as phugoid and the faster pair as short-period.

Table 6.2: Current restricted longitudinal modal-validation results for the DATCOM candidate path.

Case	Mode	$\Re(\lambda)$ [1/s]	ω_n [rad/s]	ζ	Period [s]
clean_cruise_m070	phugoid	-0.001215	0.063889	0.01902	98.36
clean_cruise_m070	short-period	-0.363103	0.652714	0.55630	11.58
clean_cruise_m075	phugoid	-0.001819	0.066811	0.02722	94.08
clean_cruise_m075	short-period	-0.425626	0.869361	0.48959	8.29
clean_cruise_m080	phugoid	-0.001723	0.064624	0.02666	97.26
clean_cruise_m080	short-period	-0.479470	1.016779	0.47156	7.01

All three cases produce two stable oscillatory mode pairs. The phugoid natural frequency and period remain in a low-frequency, lightly damped range, while the short-period mode remains faster and more strongly damped. Therefore the validation report records the marker

$$\texttt{DATCOM\_PROMOTION\_MODAL\_VALIDATED=true}. \tag{6.12}$$

This result is still limited to the restricted longitudinal model. It does not validate lateral-directional modes, actuator dynamics, engine pitching moments or a fully coupled six-degree-of-freedom linearisation.

6.5 Reference Cases

The project defines representative reference cases in `reference_cases.m`. The first case is a nominal cruise condition at approximately 35000 ft, or 10 668 m. The second case is an approach condition at approximately 3000 ft, or 914.4 m, and a negative flight-path angle. Each case stores mass in kilograms and uses the fixed project gravity value. This avoids ambiguity associated with symbolic mass-case names and keeps the trim problem tied to explicit physical parameters.

6.6 Validation Strategy

Because certified aerodynamic and engine data are not available, validation is initially qualitative and consistency-based. The following checks define the staged validation path:

- sign checks for aerodynamic derivatives and control effects,
- static stability checks in longitudinal and lateral-directional channels,
- trim residual checks for each reference case,

- open-loop response checks for elevator, aileron, rudder and throttle perturbations,
- comparison of modal behaviour with expected transport-aircraft tendencies,
- sensitivity checks for mass and configuration changes.

6.7 Linearisation Around Trim

After trim points are found, the nonlinear plant can be linearised around selected operating points:

$$\Delta\dot{\boldsymbol{x}} = \boldsymbol{A}\Delta\boldsymbol{x} + \boldsymbol{B}\Delta\boldsymbol{u}. \tag{6.13}$$

The resulting linear models will support analysis of phugoid, short-period, Dutch-roll, roll and spiral modes. The current restricted longitudinal modal check covers only the phugoid and short-period modes. A later full-plant linearisation should add lateral-directional and coupled six-degree-of-freedom modal checks before the aircraft model is used as the final NMPC validation plant.

Chapter 7

NMPC Autopilot Design

7.1 Autopilot Architecture

The autopilot is designed as a layered system. The guidance layer generates references such as altitude, airspeed, heading, flight-path angle or path error. The NMPC layer computes control commands that minimise tracking errors while respecting constraints. The actuator and propulsion layer maps commands to actual elevator, aileron, rudder and thrust states.

7.2 Optimisation Problem

At each control update the NMPC controller solves a finite-horizon optimisation problem. A representative cost function is

$$J = \sum_{k=0}^{N_p} \|\boldsymbol{y}_k - \boldsymbol{y}_{ref,k}\|_Q^2 + \sum_{k=0}^{N_c-1} \|\boldsymbol{u}_k - \boldsymbol{u}_{ref,k}\|_R^2 + \sum_{k=0}^{N_c-1} \|\Delta \boldsymbol{u}_k\|_S^2. \tag{7.1}$$

The terms in the cost function penalise tracking errors, absolute control usage and control increments. Additional soft-constraint penalties may be added for flight-envelope variables when strict feasibility is difficult to guarantee.

7.3 Constraints

The most important constraints are:

- elevator, aileron and rudder deflection limits,
- control-surface rate limits,
- thrust and throttle limits,

- bank-angle and pitch-angle limits,
- angle-of-attack and sideslip limits,
- airspeed bounds,
- vertical-speed or flight-path-angle bounds,
- load-factor bounds.

The explicit treatment of these limits is the main reason for selecting NMPC rather than a purely classical autopilot design for this project.

7.4 Reduced Prediction Models

The first implementation will use separate reduced prediction models for longitudinal and lateral-directional control. The longitudinal model will include airspeed, angle of attack, pitch rate, pitch angle, altitude and equivalent thrust. The lateral-directional model will include sideslip, roll rate, yaw rate, bank angle, heading angle and lateral path error. This separation reduces computational complexity and allows each channel to be validated independently before any integrated NMPC formulation is attempted.

7.5 Planned Autopilot Modes

The planned modes include:

- altitude hold,
- speed hold,
- vertical-speed or flight-path-angle control,
- heading select,
- simplified lateral navigation tracking,
- simplified vertical navigation tracking,
- localiser and glideslope capture in a simplified approach scenario.

Chapter 8

Simulation Studies

8.1 Purpose of Simulation Studies

The simulation studies will evaluate both the physical behaviour of the aircraft model and the performance of the NMPC autopilot. The studies should answer whether the model can be trimmed, whether its responses are qualitatively reasonable and whether the controller can track references without violating actuator or flight-envelope constraints.

8.2 Planned Scenarios

The planned scenarios include:

1. altitude hold after an initial perturbation,
2. commanded altitude change,
3. commanded airspeed change,
4. heading hold and heading change,
5. simplified lateral path tracking,
6. descent along a prescribed vertical path,
7. localiser and glideslope capture,
8. asymmetric thrust or engine-response perturbation,
9. changed aircraft mass,
10. actuator saturation and rate-limit stress test.

8.3 Evaluation Metrics

The following metrics will be used to assess simulation results:

- steady-state tracking error,
- overshoot,
- settling time,
- maximum control-surface deflection,
- maximum control-surface rate,
- maximum thrust command,
- constraint violation magnitude and duration,
- smoothness of the response,
- numerical robustness of the simulation.

8.4 Presentation of Results

Results will be presented using time histories, comparison tables and concise engineering interpretation. Each scenario should state the initial condition, aircraft configuration, mass, altitude, airspeed, controller parameters and active constraints. Plots should include reference and measured values on the same axes wherever possible.

8.5 Expected Engineering Interpretation

The interpretation will focus on trade-offs. A fast response may require aggressive control activity and can approach rate limits. A conservative response may respect constraints easily but may lead to unacceptable tracking error. The final assessment will therefore consider both tracking quality and operational realism.

Chapter 9

Conclusions and Further Work

The thesis project establishes the foundations of a nonlinear flight dynamics simulation environment for a Boeing 777-like aircraft. The current work defines the coordinate conventions, state and input vectors, geometry data, mass and inertia estimates, standard atmosphere model, first-cut aerodynamic model and validation-oriented reference cases. These elements provide the data layer required before the six-degree-of-freedom plant and NMPC controller can be integrated.

The most important modelling decision at the current stage is the separation of fixed aircraft properties from environment-dependent force quantities. Mass and inertia are stored as aircraft properties, while weight and thrust-to-weight quantities are computed using the current gravity assumption. Although the project presently uses fixed standard gravity, this interface leaves a clear path toward a future variable-gravity or higher-fidelity environment model.

Further work should focus on implementing the force and moment build-up, integrating the nonlinear equations of motion, adding actuator and engine dynamics, developing trim routines and validating the open-loop dynamic behaviour. After these steps, the longitudinal and lateral-directional NMPC controllers can be implemented and tested in the planned simulation scenarios.

The final engineering objective is not to reproduce a certified Boeing 777 simulator, but to obtain a transparent, consistent and extensible platform for studying flight dynamics and constrained autopilot design.

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Appendix A

Project Directory Structure

This appendix summarises the working project structure used during implementation.

Path	Description
`Makefile`	repository-level build helper
`B777_NMPC_Flight_Simulator.pdf`	published thesis PDF in the repository root
`data/b777_constants.m`	project-wide constants
`data/b777_geometry.m`	aircraft geometry and reference dimensions
`data/b777_mass_inertia.m`	mass envelope and inertia estimates
`data/b777_atmosphere_isa.m`	simple ISA atmosphere model
`data/aerodynamics/b777_aero_database.m`	staged aerodynamic database interface and source inventory
`data/aerodynamics/b777_aero_simple.m`	first-cut aerodynamic coefficient model
`data/aerodynamics/b777_aero_clean_crm_seed.m`	NASA CRM clean Mach-alpha grid provider
`data/aerodynamics/b777_aero_read_crm_clean_grid_csv.m`	curated NASA CRM clean grid CSV reader
`data/aerodynamics/b777_aero_read_crm_force_moment_csv.m`	NASA CRM force and moment CSV reader
`data/aerodynamics/b777_aero_read_derivative_csv.m`	source-oriented derivative CSV reader
`data/aerodynamics/b777_aero_read_mach_derivative_csv.m`	source-oriented Mach-dependent derivative candidate CSV reader
`data/aerodynamics/b777_aero_read_high_lift_csv.m`	source-oriented high-lift CSV reader
`data/aerodynamics/b777_aero_derivative_seed.m`	DATCOM-ready residual derivative data

`data/aerodynamics/b777_aero_high_lift_seed.m`	CRM-HL/DATCOM-ready high-lift increments
`data/aerodynamics/raw/nasa_crm/NTF197_TWICS_clean_grid.csv`	curated NASA CRM NTF197 clean Mach-alpha grid
`data/aerodynamics/raw/nasa_crm/NASA_wbt0_M85_CFD_orig.csv`	mirrored NASA CRM clean CFD cross-check table
`data/aerodynamics/raw/datcom/b777_like_derivative_seed.csv`	DATCOM-ready derivative seed table
`data/aerodynamics/raw/datcom/b777_like_digital_datcom_static_damping.inp`	B777-like Digital DATCOM input deck
`data/aerodynamics/raw/datcom/b777_like_derivative_datcom_candidate_m060.csv`	extracted DATCOM derivative candidate table
`data/aerodynamics/raw/datcom/mach_grid`	committed Digital DATCOM Mach-grid input decks and candidate CSV files
`data/aerodynamics/raw/datcom/mach_grid/b777_like_datcom_mach_grid_run_index.csv`	Digital DATCOM Mach-grid run index
`data/aerodynamics/raw/datcom/mach_grid/b777_like_derivative_datcom_candidate_mach_grid.csv`	consolidated Digital DATCOM Mach-grid derivative candidate table
`data/aerodynamics/raw/datcom/mach_grid/b777_like_derivative_datcom_promotion_gates.txt`	Digital DATCOM derivative promotion-gate report
`data/aerodynamics/raw/datcom/control_grid`	generated Digital DATCOM elevator and aileron control-grid decks and candidate CSV files
`data/aerodynamics/raw/datcom/control_grid/b777_like_control_derivative_datcom_candidate_mach_grid.csv`	consolidated Digital DATCOM control-derivative candidate table
`data/aerodynamics/raw/datcom/derived_gap_fill`	derived Digital DATCOM gap-fill candidate table and report

`data/aerodynamics/raw/datcom/derived_gap_fill/b777_like_derivative_gap_fill_candidate_mach_grid.csv`	consolidated derived candidate table for active-interface gaps
`data/aerodynamics/validation/trim_validation_cases.csv`	clean-cruise longitudinal trim-validation case table
`data/aerodynamics/validation/longitudinal_modal_validation_cases.csv`	clean-cruise longitudinal modal-validation case table
`data/aerodynamics/raw/high_lift/b777_like_high_lift_seed.csv`	CRM-HL/DATCOM-ready high-lift seed table
`docs/trim_validation_report.md`	clean-cruise longitudinal trim-validation report
`docs/longitudinal_modal_validation_report.md`	clean-cruise longitudinal modal-validation report
`plant/b777_aero_forces_moments.m`	aerodynamic coefficient-to-load adapter for the plant
`tools/run_digital_datcom.sh`	Digital DATCOM download, build and run helper
`tools/run_datcom_mach_grid.sh`	Digital DATCOM Mach-grid run helper
`tools/run_datcom_control_grid.sh`	Digital DATCOM control-grid run helper
`tools/generate_datcom_control_grid.py`	Digital DATCOM control-grid input deck generator
`tools/extract_datcom_derivatives.py`	Digital DATCOM output parser and candidate CSV extractor
`tools/extract_datcom_control_derivatives.py`	Digital DATCOM control-output parser and candidate CSV extractor
`tools/build_datcom_mach_grid_table.py`	Digital DATCOM Mach-grid candidate consolidation helper
`tools/build_datcom_control_grid_table.py`	Digital DATCOM control-grid candidate consolidation helper
`tools/build_datcom_gap_fill_candidates.py`	derived DATCOM gap-fill candidate builder
`tools/run_trim_validation.py`	clean-cruise longitudinal trim-validation helper
`tools/run_longitudinal_modal_validation.py`	clean-cruise longitudinal modal-validation helper

`tools/evaluate_datcom_promotion_gates.py`	Digital DATCOM candidate promotion-gate evaluator
`data/b777_configs.m`	clean, approach and landing configuration definitions
`data/reference_cases.m`	reference flight cases for trim and validation
`data/conventions.m`	units, frames, states and input conventions
`tests/features/test_project_structure.m`	project-structure feature test
`tests/features/test_modeling_conventions.m`	modelling-convention feature test
`tests/features/test_aircraft_data.m`	geometry, mass, atmosphere and reference-case feature test
`tests/features/test_aerodynamic_database.m`	staged aerodynamic-database feature test
`tests/run_feature_tests.m`	feature-test runner

Appendix B

Work Log

This appendix records the engineering workflow used during the early project stages. It is intended as a traceability aid during development and may be shortened or removed in the final submitted thesis.

Stage 1: Project Skeleton

The initial directory structure was created, including folders for aircraft data, plant dynamics, controller design, guidance logic, validation scripts, feature tests, tooling, plots, Simulink models and documentation. A LaTeX thesis draft was added so that documentation could evolve together with the implementation.

Stage 2: Modelling Conventions

The main reference frames, units, state vectors, input vectors and control-surface sign conventions were defined. The full plant state was extended to include actual engine thrust and actuator states.

Stage 3: Aircraft Data Layer

The first reusable aircraft data modules were implemented. Geometry, mass, inertia, standard atmosphere, reference cases and a simple aerodynamic coefficient model were introduced. Feature checks were updated to verify the consistency of these modules.

Stage 4: Staged Aerodynamic Database

The aerodynamic model was reorganised around a database-style interface. The initial analytical model remains available as a low-confidence seed, while the new database layer records the intended source hierarchy: public B777-like geometry and mass data, NASA CRM clean transonic aerodynamics, DATCOM derivative estimates, NASA CRM-HL high-lift data and selected OpenFOAM verification cases. The first NASA CRM CFD table at $M = 0.85$ was added as a loader cross-check, and the active clean-cruise baseline was then upgraded to a NASA CRM NTF197/TWICS Mach-alpha grid for `CL`, `CD` and `Cm`. Dedicated CSV readers were introduced so that NASA CRM force and moment data can be added as source files rather than manually copied coefficient vectors. Seed derivative and high-lift tables were separated from the main database so that DATCOM and CRM-HL data can later replace them without changing the plant interface. The derivative layer was then upgraded from passive metadata to an active DATCOM-ready residual table used for sideslip, rate and control effects inside the CRM clean-grid domain. A reproducible Digital DATCOM workflow was then added: the public PDAS archive is downloaded and compiled locally, committed B777-like clean input decks are run across a five-point Mach grid, and each output is parsed automatically into a candidate derivative table with a separate extraction report documenting selected rows, skipped blank fields and `NDM` entries. The per-Mach derivative files were then consolidated into a Mach-dependent candidate table with a grid-level source identifier, retained per-run source identifiers and a no-extrapolation interpolation policy. Promotion gates were then added so that source traceability and sign checks can pass while activation remains blocked until active-interface coverage, control derivatives and validation evidence are available. The DATCOM workflow was then extended with generated elevator and aileron control-effectiveness decks. The new control-grid runner executes `SYMFLP` and `ASYFLP` cases over the same Mach stations, extracts elevator and aileron candidates, consolidates them into a separate control-grid table and records the entries that are not direct DATCOM printouts. A derived gap-fill builder was then added for C_{D_β}, C_{Y_r}, rudder derivatives and lateral control drag, using the DATCOM candidate tables and B777-like geometry so that active-interface coverage can be checked without silently using seed values. A clean-cruise longitudinal trim-validation helper was then added for the DATCOM candidate path; three representative Mach cases pass the current force, moment and static-equivalent thrust checks. A restricted longitudinal modal-validation helper was then added for the same cases; each case produces stable phugoid and short-period mode pairs, while the DATCOM warning review remains open. The high-lift layer was then upgraded to a CRM-HL/DATCOM-ready source table containing clean, approach and landing configurations with flap, slat, gear, increment and `CLmax` metadata. The plant-facing aerodynamic load adapter was added to compute dynamic pressure, body-axis forces, aerodynamic moments and centre-of-gravity moment shifts. The aerodynamic feature test now verifies the database references, source inventory, derivative inventory, CRM grid values, DATCOM-ready residual effects, DATCOM candidate-run artifacts, trim and modal validation evidence, CRM-HL/DATCOM-ready high-lift behaviour, force signs and CG-shift logic.